

Cristian Barrera

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Summary

Machine learning and medical imaging engineer with 7+ years of experience in visiting scientist roles, academic research, industry internships, and production ML/OCT pipelines, spanning deep learning, computer vision, medical imaging, and MLOps. Strong background in Python, PyTorch, and distributed training, with a track record of shipping research-grade models and tools into real-world clinical and research workflows.

Experience

Software Developer II | Cleveland Clinic

Nov 2024 – Present

Cleveland, OH

- Lead foundation model training on 2M+ unlabeled and 600k labeled OCT volumes.
- Architected a distributed training and inference pipeline across 20+ GPUs on 10+ Linux workstations using Podman and Redis.
- Built a Flask application to orchestrate jobs, monitor status, and visualize outputs for thousands of OCT cases.
- Implemented MLflow-based experiment tracking and a model registry for foundation, fine-tuned, and RAG workflows.
- Use GitHub Copilot, Claude, and other AI coding assistants to accelerate tooling, code quality, and experimentation.

Imaging Scientist Intern | Genentech

Jun 2023 – Aug 2023

South San Francisco, CA

- Improved imaging data processing efficiency by ~30% through Python-based automation of analysis workflows.
- Developed ETL patterns for immunofluorescence imaging, integrating multiple data sources into unified research-ready datasets.

Graduate Research Scientist | Emory University

Feb 2018 – Jan 2024

Atlanta, GA

- Designed data models and visualizations for 4,000+ histopathology images for cancer research and biomarker discovery.
- Developed machine learning models for patient outcomes and immunotherapy response, contributing to multiple peer-reviewed publications.
- Built scalable deep learning and traditional ML pipelines for biomedical imaging and clinical data, integrating with HPC and cloud resources.

Biomedical Engineering Contractor | National University of Colombia

Jan 2017 – Dec 2019

Bogotá, Colombia

- Developed a deep learning-based face recognition tool to track children in daycare and ICBF programs, including strategies for retraining models as children grow.
- Designed data collection and evaluation workflows to monitor system accuracy, coverage, and false alerts in real-world environments.
- Supported the design and testing of a surface EMG system for pregnant patients to capture maternal and fetal heart rates over 9-month gestation.
- Led bench testing of EMG hardware (including electrode/prong setups) to ensure reliable heartbeat capture for subsequent signal separation using linear discriminant analysis.

Visiting Scientist (J-1 Short-Term Scholar) | Case Western Reserve University Nov 2017 – Jan 2020
Cleveland, OH

- Conducted short-term visiting appointments as research staff in biomedical engineering.
- Led and supported imaging and data analysis workflows, collaborating with faculty and clinicians on histopathology imaging projects.
- Applied deep learning and signal processing methods to experimental datasets, providing reproducible code and documentation for lab use.

Projects

Distributed OCT Foundation Model Pipeline

- End-to-end pipeline for training and deploying OCT foundation models on 2M+ unlabeled and 600k labeled samples across 20+ GPUs and 10+ machines using Podman containers, Redis coordination, MLflow tracking, and a Flask front end.

AI Agent Reader for Quality Control

- Vision and language model-based agent to verify, correct, and summarize workflow outputs before QC review, integrated into containerized pipelines with automated logging.

Skills

Programming: Python, R

ML / DL: PyTorch, TensorFlow, computer vision, deep learning, foundation models, representation learning, model evaluation and validation

MLOps / Data: Flask, Docker, Podman, Redis, MLflow, data pipelines, ETL, experiment tracking, model registry, distributed training, Linux, HPC, AWS, Azure

Domains: Medical imaging, OCT, histopathology, biomedical signal processing, clinical data, ophthalmology

Education

Ph.D. Biomedical Engineering | Georgia Tech & Emory University Aug 2020 – May 2025
Atlanta, GA

- Focus: medical imaging, histopathology, ophthalmology, deep learning-based analysis.

Master of Engineering, Biomedical Engineering | National University of Colombia Jan 2017 – Dec 2019
Bogotá, Colombia

- Focus: medical image processing, machine learning, genomics.

Bachelor of Engineering, Electronics | South Colombian University Jan 2011 – Dec 2015
Neiva, Colombia

- Focus: digital electronics, signal processing, telecommunications.